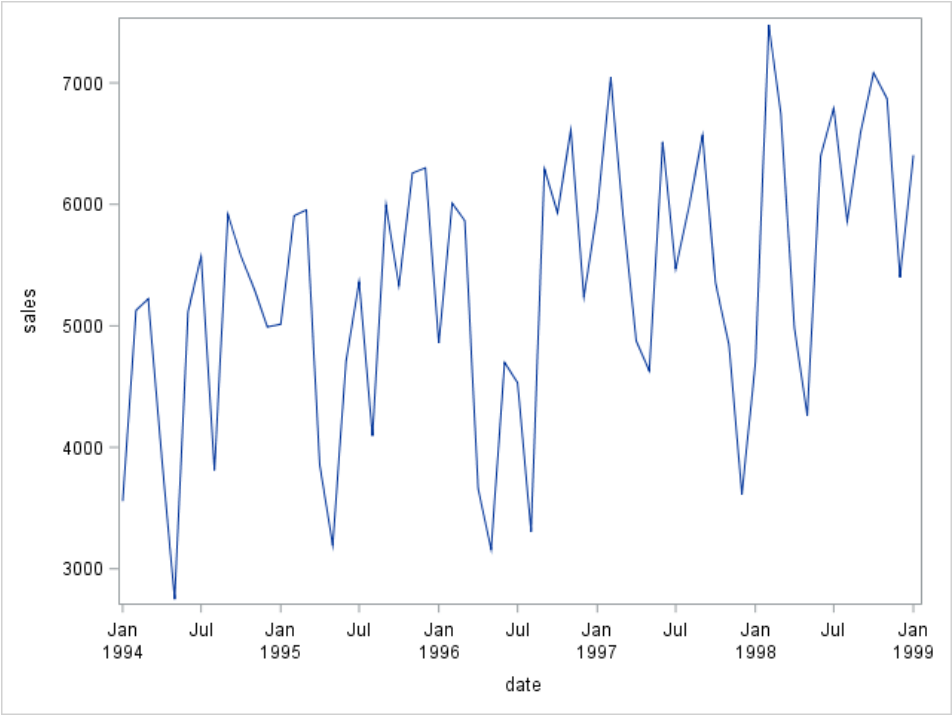
Note 5 Exponential Smoothing

Exponential Smoothing is a technique for smoothing a time series using exponential filter. This technique is widely used in forecasting time series. Exponential smoothing is especially useful when the series is relatively short (small sample).

1 Simple Exponential Smoothing

The simplest of the exponentially smoothing methods is **simple exponential smoothing**, also known as **single exponential smoothing**. This method is appropriate for forecasting series with no clear trend or seasonal patterns.

Example 1.1. The monthly shoes sales from Jan 1994 to Jan 1999 are shown below.



It time series, it makes more sense to assign larger weights to recent observations than to observations from the distant past. Exponential smoothing is based on this concept.

Component form representations of exponential smoothing methods comprise a forecast equation and a smoothing equation for each of the components included in the method:

$$\text{Forecast equation: } \hat{X}_{t+h|t} = \ell_t$$

$$\text{Level equation: } \ell_t = \alpha X_t + (1-\alpha)\ell_{t-1}$$

Remarks:

- Simple exponential smoothing has a "flat" forecast function:

  $$\hat{X}_{t+h|t} = \hat{X}_{t+1|t} = \ell_t \text{ for all } h = 2, 3, \cdots$$

  That is, all forecasts take the same value, equal to the last level component. Remember that these forecasts will only be suitable if the time series has no trend or seasonal component.

- The component form of simple exponential smoothing is not particularly useful, but it will be the easiest form to use when we start adding other components (e.g., trend, seasonality).

In practice, the exponential smoothing method requires us to select the value of the smoothing parameter α.

- In some cases, the smoothing parameters are chosen based on previous experience.
- A more objective way is estimate the smoothing parameters from the observed data. Least square estimation is a common choice.

2 Holt's Linear Trend Method

The **Holt's method** is an extension of the simple exponential smoothing that allows the forecasting of data with a trend.

Holt's linear trend method involves a forecast equation and two smoothing equations (one for the level and one for the trend):

$$\text{Forecast equation: } \hat{X}_{t+h|t} = \ell_t + hb_t$$

$$\text{Level equation: } \ell_t = \alpha X_t + (1-\alpha)(\ell_{t-1} + b_{t-1})$$

$$\text{Trend equation: } b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}$$

where ℓ_t denoted an estimate of the level of the series at time t, b_t denotes an estimate of the trend of the series at time t, α is the smoothing parameter with $0 \leq \alpha \leq 1$, and β is the smoothing parameter for the trend with $0 \leq \beta \leq 1$.

3 Holt-Winters Seasonal Method

Holt-Winters method extends Holt's method to capture seasonality.

The Holt-Winters seasonal method comprises the forecast equation and three smoothing equations:

$$\text{Forecast equation: } \hat{X}_{t+h|t} = \ell_t + hb_t + S_{t+h-T(k+1)}$$

$$\text{Level equation: } \ell_t = \alpha(X_t - S_{t-T}) + (1-\alpha)(\ell_{t-1} + b_{t-1})$$

$$\text{Trend equation: } b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}$$

$$\text{Seasonality equation: } S_t = \gamma(X_t - \ell_t) + (1-\gamma)S_{t-T}$$

where T is the period of the series, and k is the integer part of $(h-1)/T$.